

N 1

$$I_{k,j} = \left[\frac{j-1}{k}; \frac{j}{k} \right] \Rightarrow A_1, A_2, \dots, A_n - \text{разбиение } [0, 1] :$$

$$F_n(x) = I_{A_n}(x)$$

$$\boxed{A_1 = [0, 1], \quad A_2 = \left[0, \frac{1}{2}\right], \quad A_3 = \left[\frac{1}{2}, 1\right], \dots}$$

$$\int_0^1 F_n(x) dx = |A_n| = \frac{1}{k}; \quad n \rightarrow \infty \Rightarrow \frac{1}{k} \rightarrow 0$$

$$\Rightarrow \limsup_{n \rightarrow \infty} \int_0^1 F_n(x) dx = 0$$

$$\limsup_{n \rightarrow \infty} F_n(x) = 1, \text{ т.к. } \forall N \in \mathbb{N} \quad \exists n \geq N : F_n(x) = 1$$

$$\Rightarrow \int_0^1 \limsup_{n \rightarrow \infty} F_n(x) dx = 1 > 0 = \limsup_{n \rightarrow \infty} \int_0^1 F_n(x) dx$$

N 2

a)

$$f(x) = \sqrt{x}$$

$$F(x) = F(0) + \int_0^x \frac{1}{2\sqrt{t}} dt; \quad \frac{1}{2\sqrt{t}} \in L^1(0, 1), \text{ т.к. } \int_0^1 \left| \frac{1}{2\sqrt{t}} \right| dt < \infty$$

$$\Rightarrow F(x) - \text{абс. непр.}$$

$$y = 0 : \quad \frac{|\sqrt{x} - 0|}{|x - 0|} = \frac{1}{\sqrt{x}} \rightarrow \infty \quad (\sqrt{x} \rightarrow 0) - \text{не липшицева}$$

б)

$$F(x) = \begin{cases} 0, & x \in [0; \frac{1}{2}) \\ 1, & x \in [\frac{1}{2}; 1] \end{cases}$$

N 3

a)

$$u = (\cos \theta, \sin \theta)$$

$$\lim_{t \rightarrow 0} \frac{F(t \cos \theta, t \sin \theta) - F(0, 0)}{t} = \cos^3 \theta \Rightarrow |\cos^3 \theta| \leq 1$$

б) если $\gamma(t) \neq (0,0)$, то $F(\gamma(t))$ — непр. дифф. если $\gamma(t_0) = (0,0)$, то $\gamma'(t_0) = \lim_{h \rightarrow 0} \frac{\gamma(t_0+h) - \gamma(t_0)}{h}$

$$h\gamma'(t_0) + o(h) = \gamma(t_0 + h) - \gamma(t_0) = \gamma(t_0 + h)$$

$$F(\gamma(t_0 + h)) = F(h\gamma'(t_0) + o(1)) = hF(\gamma'(t_0) + o(1))$$

$$\frac{F(\gamma(t_0 + h)) - F(\gamma(t_0))}{h} = \frac{F(\gamma(t_0 + h))}{h} = F(\gamma'(t_0) + o(1)) \rightarrow F(\gamma'(t_0))$$

$$\Rightarrow (F(\gamma(t_0)))' = F(\gamma'(t_0))$$

$$(F(\gamma(t)))' = \nabla F(\gamma(t)) \cdot \gamma'(t)$$

$$\nabla F(\gamma(t_0+h)) = \nabla F\left(\frac{\gamma(t_0+h)}{|\gamma(t_0+h)|}\right) = \nabla F\left(\frac{h\gamma'(t_0) + o(h)}{|h\gamma'(t_0) + o(h)|}\right) = \nabla F\left(\frac{\gamma'(t_0) + o(1)}{|\gamma'(t_0) + o(1)|}\right)$$

$$= \nabla F(\gamma'(t_0) + o(1)) \rightarrow \nabla F(\gamma'(t_0)) \text{ при } t \rightarrow t_0, \gamma'(t_0) \neq 0$$

$$|\nabla F(\gamma(t))| \leq M \Rightarrow |(F(\gamma(t)))'| \leq M|\gamma'(t)| \rightarrow 0 = \gamma'(t_0),$$

при $t \rightarrow t_0, \gamma'(t_0) = 0$

в)

$$dF(x, y) = F'_x(x, y)dx + F'_y(x, y)dy = dx$$

Опр. дифференцируемости в т. $(0,0)$:

$$F(0 + h, 0 + k) = F(0, 0) + dF(0, 0)(h, k) + o\left(\sqrt{h^2 + k^2}\right) = h + o\left(\sqrt{h^2 + k^2}\right)$$

$$\Rightarrow (!) : F(h, k) - h = o\left(\sqrt{h^2 + k^2}\right)$$

$$F(t, t) = \frac{t}{2}$$

$$F(t, t) - t = -\frac{t}{2} \Rightarrow \frac{F(t, t) - t}{\sqrt{t^2 + t^2}} = \frac{-\frac{t}{2}}{|t|\sqrt{2}} = -\frac{\text{sgn}(t)}{2\sqrt{2}} \not\rightarrow 0,$$

при $t \rightarrow 0$

$$\Rightarrow F \text{ не дифф. в } (0, 0)$$

№ 4

$$\|A\| := \sup_{\|x\| \leq 1} \|Ax\|$$

$$(Ax)_i = \sum_{j=1}^n a_{ij}x_j$$

$$\|Ax\|^2 = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij}x_j \right)^2$$

$$\|Ax\|^2 \leq \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij}^2 \right) \|x\|^2 = \|x\|^2 \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2 \right)$$

$$\text{при } \|x\| \leq 1 : \quad \|Ax\| \leq \left(\sum a_{ij}^2 \right)^{\frac{1}{2}}$$

$\Downarrow$

$$\|A\| \leq \left(\sum a_{ij}^2 \right)^{\frac{1}{2}}$$

№ 5

$$y_0 \in I; \quad \Phi(y) = \int_E f(x, y) dx$$

$$(!) : \Phi(y) \text{ — дифф-ма в т. } y_0 \text{ и } \frac{d\Phi(y)}{dy} = \int_E \frac{\partial f(x, y)}{\partial y} dx$$

$$h \neq 0, \quad y_0 + h \in I$$

$$\frac{\Phi(y_0 + h) - \Phi(y_0)}{h} = \int_E \frac{f(x, y_0 + h) - f(x, y_0)}{h} dx$$

$$g_h(x) = \frac{f(x, y_0 + h) - f(x, y_0)}{h} = \frac{\partial f}{\partial y}(x, \xi), \quad \xi \in (y_0, y_0 + h)$$

$$\left| \frac{\partial f}{\partial y}(x, \xi) \right| \leq F(x), \quad F(x) \text{ — мажоранта}$$

$$\lim_{h \rightarrow 0} g_h(x) = \frac{\partial f}{\partial y}(x, y_0)$$

по т. Лебега о мажорир. сх.:

$$\lim_{h \rightarrow 0} \int_E g_h(x) dx = \int_E \lim_{h \rightarrow 0} g_h(x) dx$$

$$\Rightarrow \lim_{h \rightarrow 0} \int_E \frac{f(x, y_0 + h) - f(x, y_0)}{h} dx = \int_E \frac{\partial f(x, y_0)}{\partial y} dx$$

$$\frac{d}{dy} \int_E f(x, y_0) dx = \int_E \frac{\partial f(x, y_0)}{\partial y} dx$$